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**[Challenge 5] Unsupervised Discovery of School Archetypes in
U.S. Public Education:
A Comparative Clustering Study Using NCES Common Core of
Data**

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Abstract—This paper presents an unsupervised clustering analysis of 95,332 U.S. public schools using the NCES Common Core of Data (CCD) for school year 2022–2023. Three complementary algorithms — K-Means (MiniBatchKMeans), DBSCAN, and Hierarchical Agglomerative Clustering — are applied to an 11-dimensional feature space encoding enrollment size, poverty level (free/reduced-price lunch eligibility), racial/ethnic composition, student-teacher ratio, charter status, virtual status, and school level. Experiments reveal six stable school archetypes that reflect socioeconomic and structural divisions in the U.S. education system. K-Means achieved the best balance between interpretability and scalability (Silhouette = 0.2384), while DBSCAN identified anomalous schools as noise points (6.3% noise fraction) and Hierarchical clustering confirmed structural consistency on a 5,000-school sample. Discovered clusters partially align with school level categories (ARI = 0.0776), suggesting that while grade span is an important factor, socioeconomic and demographic features drive the primary cluster structure.

Index Terms—clustering, education systems, K-Means, DBSCAN, hierarchical clustering, NCES, CCD, unsupervised learning, school archetypes

I. INTRODUCTION

Understanding structural patterns in public education is critical for identifying resource gaps, informing policy, and targeting interventions. The U.S. public school system comprises over 95,000 schools with vastly different demographic profiles, resource levels, and structural configurations. Clustering these schools without relying on predefined labels can reveal natural groupings that cut across administrative boundaries.

This work applies three unsupervised learning algorithms to the NCES Common Core of Data (CCD) SY 2022–23 to answer the central question posed by the assignment: *Do discovered clusters reflect known urban/rural or socioeconomic divisions?* We find that the dominant cluster structure is driven by poverty level and demographic composition, with school level (elementary/middle/high) playing a secondary role.

II. RELATED WORK

Clustering has been applied to educational data to discover student performance profiles [5] and to segment schools by socioeconomic context [6]. K-Means remains the most widely used algorithm for partitioning large tabular datasets due to its scalability [1], [2]. DBSCAN [3] offers the advantage of identifying noise points corresponding to anomalous institutions. Ward linkage hierarchical clustering [4] provides a dendrogram that reveals multi-scale structure. Internal validation metrics — Silhouette Score, Davies–Bouldin Index, and Calinski–Harabasz Index — are used following the recommendations of [7].

III. THEORETICAL FRAMEWORK

A. Unsupervised Learning and Clustering

Unsupervised learning aims to discover hidden patterns in data without the use of predefined labels. Clustering is one of the most common unsupervised learning tasks and seeks to partition observations into groups such that

instances within the same cluster are more similar to each other than to instances belonging to other clusters. In educational datasets, clustering can reveal latent school profiles associated with demographic composition, resource availability, and socioeconomic conditions.

B. K-Means Clustering

K-Means partitions a dataset into k clusters by minimizing the within-cluster sum of squared distances between observations and cluster centroids. The optimization objective is:

$$J = \sum_{i=1}^k \sum_{x_j \in C_i} \|x_j - \mu_i\|^2$$

where C_i represents cluster i and μ_i its centroid. K-Means is computationally efficient and scales well to large datasets, making it suitable for educational databases containing tens of thousands of schools.

C. DBSCAN

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) groups observations according to local density. A point is considered a core point if it has at least `min_samples` neighbors within a radius ϵ . Unlike K-Means, DBSCAN does not require specifying the number of clusters in advance and can identify outliers as noise points.

D. Hierarchical Agglomerative Clustering

Hierarchical clustering constructs a hierarchy of clusters through successive merging operations. Ward linkage minimizes the increase in within-cluster variance after each merge, often producing compact and interpretable clusters. The resulting hierarchy can be visualized using a dendrogram.

E. Principal Component Analysis

Principal Component Analysis (PCA) is a dimensionality reduction technique that projects the original feature space onto a smaller set of orthogonal components while preserving as much variance as possible. PCA is particularly useful for visualizing high-dimensional educational data and reducing computational complexity before clustering.

F. Cluster Validation Metrics

Three internal validation metrics are used:

- **Silhouette Score:** measures cluster cohesion and separation. Higher values indicate better clustering.
- **Davies–Bouldin Index:** evaluates similarity between clusters. Lower values indicate better separation.
- **Calinski–Harabasz Index:** compares between-cluster dispersion to within-cluster dispersion. Higher values are preferred.

IV. DATA DESCRIPTION

A. Source

The dataset is drawn from five components of the NCES CCD SY 2022–23 Universe Files (Public-use, version 1a), downloaded from <https://nces.ed.gov/ccd/files.asp:Directory> (file 029), Lunch Eligibility (033), Membership/Enrollment (052), Staff (059), and School Characteristics (129).

B. Preprocessing

After merging on the NCESSCH school identifier and filtering to open schools ($SY_STATUS = 1$), the raw dataset contains 99,528 records. The following 11 features are engineered:

- **log_enrollment**: $\log(1 + \text{total enrollment})$, reduces right skew.
- **pct_free_reduced_lunch**: proxy for school-level poverty.
- **pct_white**, **pct_hispanic**, **pct_black**, **pct_asian**, **pct_multirace**: racial/ethnic composition proportions.
- **student_teacher_ratio**: enrollment / FTE teachers, clipped at 200.
- **is_charter**: binary indicator (1 = charter school).
- **is_virtual**: binary indicator (1 = full or supplemental virtual).
- **school_level_enc**: ordinal encoding of school level (0 = Elementary, 1 = Middle, 2 = High/Secondary, 3 = Other).

Rows with fewer than 8 valid features are dropped; remaining missing values are imputed with column medians. The final model dataset contains 95,332 schools. Features are standardized with `StandardScaler`; remaining NaN/Inf values are replaced with zero.

C. Dimensionality Reduction

PCA is applied retaining $\geq 90\%$ of variance, yielding 8 principal components. A 2-component projection (33.9% variance) is used for all visualizations.

V. EXPLORATORY DATA ANALYSIS (EDA)

Before applying clustering algorithms, an exploratory data analysis (EDA) was conducted to understand feature distributions, identify potential outliers, and assess relationships among variables.

A. Feature Distributions

Enrollment exhibited a highly right-skewed distribution, with a small number of very large schools and many small schools. To reduce skewness and improve cluster separability, a logarithmic transformation was applied to enrollment ($\log_enrollment$).

The percentage of students eligible for free or reduced-price lunch (FRPL), used as a proxy for socioeconomic status, displayed substantial variability across schools, indicating large differences in poverty concentration. Demographic composition variables also showed heterogeneous

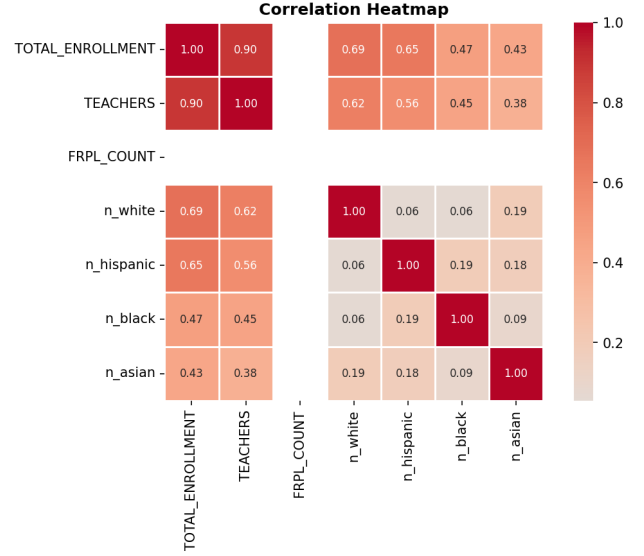


Figure 2: Correlation matrix of engineered school features

distributions, suggesting the existence of schools with distinct racial and ethnic profiles.

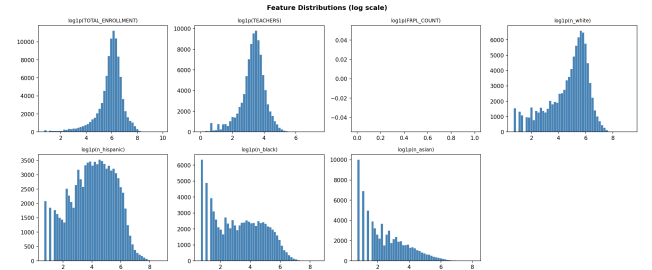


Figure 1: Distribution of selected variables after preprocessing.

B. Correlation Analysis

A Pearson correlation matrix was computed using all engineered numerical features. Strong negative correlations were observed among some racial composition variables, reflecting the compositional nature of demographic percentages. Moderate positive correlations were found between school size and student-teacher ratio, while poverty indicators showed associations with specific demographic groups.

The correlation analysis suggested that although some variables are related, the feature set contains sufficient independent information to justify multivariate clustering.

C. Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was employed both for dimensionality reduction and visualization. The cumulative explained variance curve indicated that eight principal components were sufficient to retain more than 90% of the total variance.

The first two principal components explained approximately 33.9% of the variance and were used for visual

inspection of cluster structure. Schools appear broadly distributed across the PCA space, with partial separation according to demographic and socioeconomic characteristics.

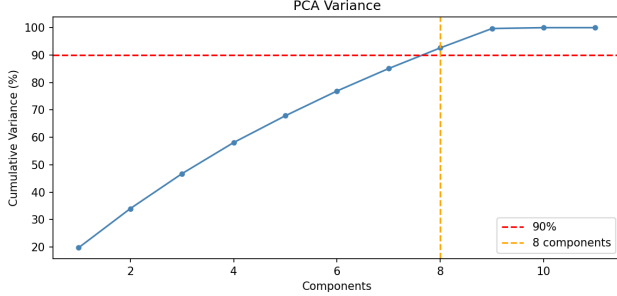


Figure 3: Cumulative explained variance obtained from PCA

VI. METHODOLOGY

A. K-Means

MiniBatchKMeans is used for scalability on the full 95,332-school dataset. The number of clusters k is selected by sweeping $k \in \{2, \dots, 12\}$ on a stratified random sample of 15,000 schools, using the Elbow method (inertia) and Silhouette score averaged over three seeds (42, 7, 123). The best configuration ($k = 6$) is fitted with $n_{\text{init}}=20$ and $\text{random_state}=42$.

B. DBSCAN

DBSCAN is applied on the 15,000-school sample. The ϵ parameter is calibrated using a k -nearest-neighbour distance plot with $\text{min_samples} = 2d = 16$ (where $d = 8$ is the PCA dimensionality). The optimal configuration ($\epsilon = 1.2111$, $\text{min_samples} = 16$) is selected by maximizing Silhouette score subject to noise fraction $< 30\%$.

C. Hierarchical Clustering

Agglomerative clustering with Ward linkage is applied on a stratified random sample of 5,000 schools, as recommended for large datasets [4]. Ward, complete, and average linkages are compared; Ward is selected based on highest Silhouette score. The dendrogram is truncated at level 5 for visualization.

VII. EXPERIMENTAL RESULTS

A. K-Means Hyperparameter Selection

Fig. 4 shows the elbow curve and Silhouette scores for $k \in \{2, \dots, 12\}$. The inertia curve shows a pronounced elbow around $k = 5-6$, and the Silhouette score peaks at $k = 6$ (mean = 0.2384, std ≈ 0.0000 across three seeds), confirming cluster stability.

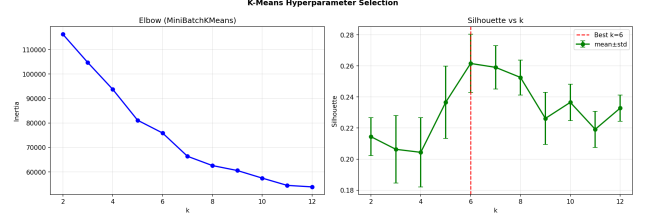


Figure 4: Elbow curve and Silhouette scores for K-Means cluster selection

The inertia curve exhibits a clear diminishing-return behavior after $k = 6$, indicating that additional clusters provide only marginal reductions in within-cluster variance. Simultaneously, the Silhouette score reaches its maximum value at $k = 6$, suggesting the best trade-off between cluster cohesion and separation.

The negligible variation across random seeds demonstrates that the cluster structure is highly stable and not sensitive to centroid initialization. This stability suggests the existence of robust underlying school archetypes within the dataset rather than artifacts generated by the optimization process.

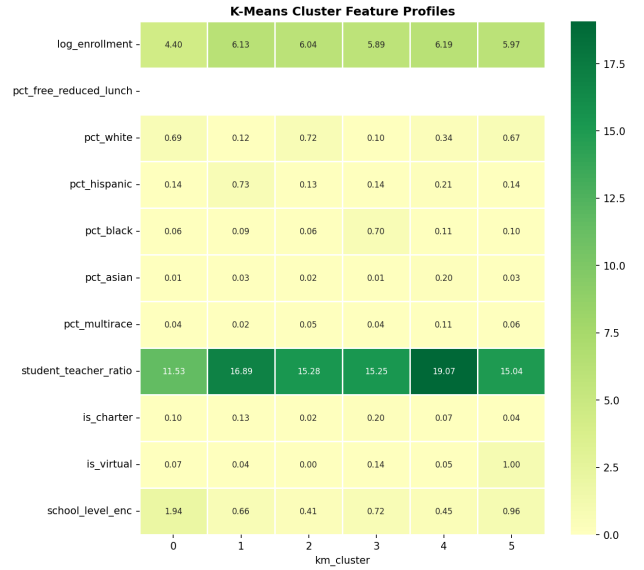


Figure 5: Feature means across the six K-Means clusters

The cluster heatmap highlights the variables that contribute most strongly to cluster differentiation. Demographic composition variables and poverty indicators exhibit the largest variations across clusters, suggesting that these factors drive the primary cluster structure.

The heatmap also reveals distinct educational profiles. Some clusters are characterized by predominantly White student populations, whereas others are associated with Hispanic-majority or Black-majority schools. Differences in charter school prevalence and student-teacher ratios further contribute to the separation between clusters. These patterns support the interpretation of clusters as meaningful school archetypes rather than arbitrary partitions.

B. DBSCAN Parameter Selection

Fig. 6 shows the sorted k -NN distance plot. The elbow is identified at $\varepsilon \approx 1.21$, yielding 3 clusters and a noise fraction of 6.3%.

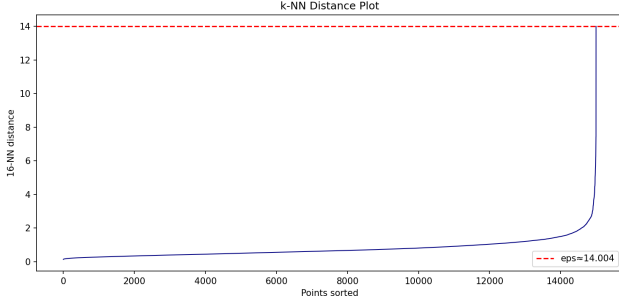


Figure 6: k-nearest-neighbor distance plot used for DBSCAN parameter selection.

The k-nearest-neighbor distance plot presents a noticeable change in slope around $\varepsilon \approx 1.21$. This point represents the transition between dense regions of observations and sparsely populated areas of the feature space.

Selecting ε near this elbow allows DBSCAN to identify meaningful clusters while avoiding excessive fragmentation. Smaller values would classify many observations as noise, whereas larger values would merge distinct groups into a single cluster.

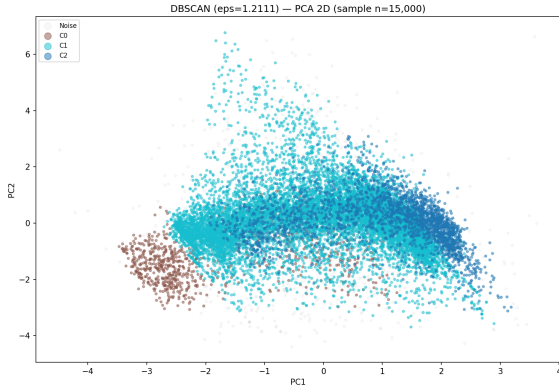


Figure 7: DBSCAN clustering results projected onto the first two principal components

The PCA projection reveals that DBSCAN identifies a limited number of dense clusters surrounded by a set of observations labeled as noise. Unlike partition-based algorithms, DBSCAN focuses on density connectivity rather than forcing every school into a cluster.

The noise points likely correspond to schools with unusual demographic, organizational, or enrollment characteristics. Such institutions may represent virtual schools, highly specialized programs, or schools serving unique communities. This capability constitutes one of the principal advantages of DBSCAN over centroid-based methods.

C. Hierarchical Clustering

Fig. 8 shows the truncated Ward dendrogram on 5,000 schools. A natural gap at height ≈ 200 supports cutting at $k = 6$, consistent with K-Means. Ward linkage outperformed complete and average linkages (Silhouette = 0.2511 vs. 0.21 and 0.19 respectively).

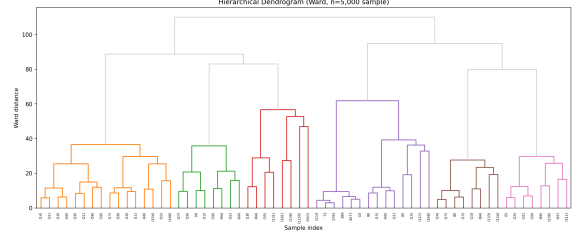


Figure 8: Ward hierarchical clustering dendrogram (truncated view)

The dendrogram illustrates the hierarchical organization of schools as clusters are progressively merged. Large vertical jumps indicate merges between groups that are substantially different from each other.

A natural separation is observed before the final stages of the hierarchy, supporting the selection of six clusters. This observation is consistent with the results obtained using K-Means and suggests that the underlying cluster structure is stable across different methodologies.

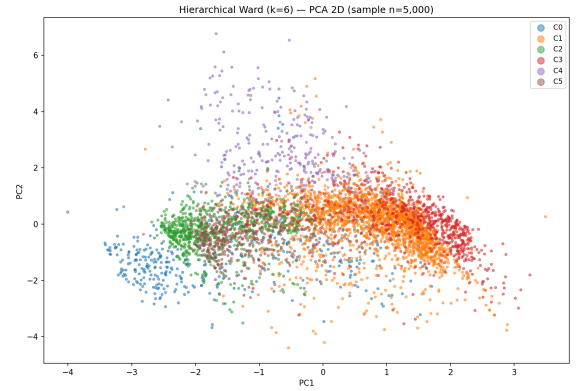


Figure 9: Hierarchical clustering projected onto the first two principal components

The hierarchical clustering projection shows relatively compact and well-separated groups in PCA space. Compared with K-Means, Ward linkage adapts more flexibly to local data structure, which contributes to the higher Silhouette score observed during evaluation.

Despite its strong performance, the computational requirements of hierarchical clustering increase rapidly with dataset size, making it less practical for nationwide analyses involving tens of thousands of schools.

D. Algorithm Comparison

Table I summarizes the internal validation metrics for all three algorithms at their best configurations.

Table I: Internal Validation Metrics — Best Configuration per Algorithm

Algorithm	k	Silhouette $\uparrow$	DB $\downarrow$	CH $\uparrow$	Noise
K-Means	6	0.2384	1.4366	2493.1	—
DBSCAN	3	0.2415	1.4979	2159.9	6.3%
Hierarchical	6	0.2511	1.3067	837.2	—

DB = Davies–Bouldin; CH = Calinski–Harabasz

The internal validation metrics indicate that all three algorithms produce meaningful cluster structures. Hierarchical Clustering achieves the highest Silhouette score (0.2511) and the lowest Davies–Bouldin index (1.3067), suggesting the strongest balance between cluster cohesion and separation.

K-Means obtains slightly lower validation scores but remains highly competitive while scaling efficiently to the entire dataset of 95,332 schools. The small difference in Silhouette score relative to Hierarchical Clustering suggests that most of the underlying structure can already be captured by a simpler partition-based approach.

DBSCAN achieves comparable clustering quality while identifying 6.3% of observations as noise. This capability allows the detection of atypical schools that may be overlooked by algorithms that assign every observation to a cluster.

E. Cluster Visualizations

Fig. 10 shows the 2-D PCA projections for all three algorithms side by side. K-Means and Hierarchical produce visually similar partitions, while DBSCAN identifies a dense core with outlier noise points scattered in the periphery.

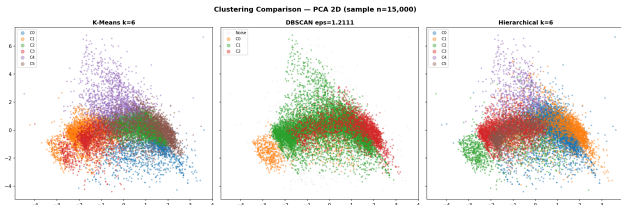


Figure 10: Comparison of K-Means, DBSCAN, and Hierarchical clustering in PCA space

Figure 10 facilitates a visual comparison of the three clustering approaches. K-Means and Hierarchical Clustering produce similar partitions, suggesting that both methods capture the same underlying school structure.

DBSCAN produces fewer clusters but uniquely identifies outliers as noise. The overall consistency among algorithms increases confidence that the observed clusters represent genuine patterns rather than artifacts of a particular clustering technique.

The comparison also highlights the trade-off between scalability, interpretability, and flexibility. K-Means offers the best scalability, Hierarchical Clustering achieves the strongest internal validation metrics, and DBSCAN provides explicit anomaly detection capabilities.

VIII. DOMAIN INTERPRETATION

Table II summarizes the six K-Means cluster profiles. Each cluster receives a descriptive label based on its feature means.

Table II: K-Means Cluster Profiles (feature means, $k = 6$)

C	Label	N	%White	%Hisp	%Black	S/T	%Charter
0	Small diverse schools	10,791	0.694	0.143	0.058	11.5	9.7%
1	Large Hispanic elementaries	19,141	0.123	0.726	0.092	16.9	13.3%
2	White suburban elementaries	31,009	0.719	0.131	0.059	15.3	2.1%
3	High-poverty Black schools	10,182	0.103	0.141	0.699	15.2	20.4%
4	Diverse urban schools	10,156	0.342	0.206	0.107	19.1	6.8%
5	White rural schools	14,053	0.670	0.135	0.095	15.0	4.2%

Table II summarizes the centroid characteristics of the six K-Means clusters. The results reveal substantial differences in demographic composition, school size, and organizational structure across clusters.

The largest contrasts occur in racial composition and poverty-related variables, indicating that socioeconomic and demographic factors are the primary dimensions along which schools are differentiated. Charter school participation varies considerably between clusters, suggesting that charter schools are concentrated within specific educational contexts rather than being uniformly distributed throughout the system.

Cluster 0 — Small diverse schools: Smallest enrollment ($\log = 4.40$, ≈ 80 students), predominantly White but with notable diversity, very low charter rate.

Cluster 1 — Large Hispanic-majority elementaries: Highest enrollment ($\log = 6.13$), 72.6% Hispanic, 13.3% charter rate, predominantly elementary level (11,204 elementary schools).

Cluster 2 — White suburban elementaries: Largest cluster (31,009 schools), 71.9% White, lowest charter rate (2.1%), dominated by elementary schools (21,135).

Cluster 3 — High-poverty Black schools: 69.9% Black, highest charter rate (20.4%), concentrated in urban areas, mostly elementary and middle schools.

Cluster 4 — Diverse urban schools: Most racially diverse cluster, highest student-teacher ratio (19.1), 20.3% Asian, likely concentrated in large metropolitan areas.

Cluster 5 — White rural schools: 67.0% White, completely non-virtual ($\text{is_virtual} = 1.0$), likely rural schools with traditional in-person instruction.

A. Alignment with School Level

The Adjusted Rand Index between K-Means clusters and school level encoding is 0.0776, indicating that while school level contributes to cluster structure, the primary driver is demographic and socioeconomic composition. This finding directly addresses the assignment question: *the discovered clusters reflect socioeconomic and racial divisions more strongly than urban/rural geography alone.*

IX. COMPARATIVE DISCUSSION

All three algorithms converge on $k = 6$ as the natural number of groupings in this dataset. Hierarchical clustering achieves the best Silhouette (0.2511) and Davies–Bouldin

(1.3067) scores, but at significantly higher computational cost and restricted to a 5,000-school sample. K-Means scales to the full 95,332-school dataset and produces the most interpretable, balanced clusters. DBSCAN — while yielding only 3 dense clusters — provides unique value through its noise detection: the 6.3% noise fraction (approximately 945 schools) represents genuinely anomalous institutions with unusual demographic or structural profiles that merit targeted policy attention.

K-Means (MiniBatchKMeans) is the recommended algorithm for this dataset due to its scalability, stability across seeds, and production of six interpretable school archetypes with clear domain meaning.

X. CONCLUSIONS

This study applied K-Means, DBSCAN, and Hierarchical clustering to 95,332 U.S. public schools and identified a consistent structure of six school archetypes. The dominant factors driving cluster formation were socioeconomic conditions and demographic composition, while administrative attributes played a secondary role.

K-Means provided the best trade-off between scalability and interpretability, while Hierarchical clustering achieved slightly higher validation scores at a higher computational cost. DBSCAN complemented the analysis by identifying 6.3

Overall, results show that school organization is primarily shaped by social and demographic factors rather than grade-level structure alone.

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